Kannan Appiya Santharam — Executive Portfolio

TECHNICAL ARCHITECTURE, HANDOVER & RECRUITER INTERVIEW PREPARATION GUIDE

Candidate Name:	Kannan Appiya Santharam	**Target Location:**	Dubai, UAE ■■
Current Role:	Senior Lead Software Engineer (10.5+ Yrs)	**Relocation Notice:**	60 Days (Visa Required)
Target Roles:	Lead Software Engineer / Engineering Manager	**Production Stack:**	React · TypeScript · Vite · Vercel
Email:	as.kannan4@gmail.com	**GitHub:**	github.com/kannan-santharam

1. System Architecture & High-Level Design

The portfolio is an enterprise-grade, AI-native single-page web application engineered with modern React and TypeScript. It features real-time HTTP Streamable token delivery powered by Google Gemini 1.5 Flash LLM, backed by serverless security proxying and full APM observability through Langfuse.

Component Layer	Technology & Pattern	Key Architectural Role
Frontend Core	React, TypeScript, Tailwind CSS, Vite	High-performance responsive UI, zero-lag theme system, mobile optimization.
AI Engine & Proxy	Serverless Endpoint (api/chat.ts)	Hides GEMINI_API_KEY on server, streams token chunks over HTTP POST.
Observability (APM)	Langfuse JS SDK v3 (Server-side)	Tracks traces & LLM generation metrics (Time-To-First-Token, output tokens).
Local Fallback Engine	Dynamic Dossier Search Engine	Client-side fallback regex search engine if server proxy or API key is offline.
Deployment & CDN	Vercel Serverless & Edge Network	Automated CI/CD build pipelines triggered on git push to main branch.

2. Frontend to Backend Communication & LLM Streaming

Real-Time HTTP Streamable Token Delivery (fetch + ReadableStream):
Unlike legacy applications that wait for the complete LLM response before rendering or use heavy WebSockets, this portfolio uses native Web Streams over chunked HTTP POST responses. The client invokes fetch('/api/chat') and reads tokens incrementally using ReadableStream.getReader(), reducing perceived latency (TTFT) to under 80ms.

Serverless Security Proxying (api/chat.ts):
The Gemini API key (GEMINI_API_KEY) is stored strictly as a server-side environment variable. The client browser never

sees or handles raw credentials. The serverless function streams text chunks directly back to the client while sanitizing output.

Langfuse APM Tracing & Telemetry:
Every recruiter query creates an active trace in Langfuse Cloud (traceId) containing LLM generation parameters, prompt token counts, latency metrics, and error logs for full operational visibility.

3. Codebase Directory Map & Key Modules

- **api/chat.ts** — Vercel Serverless endpoint, streams Gemini tokens & sends Langfuse telemetry.
- **src/components/Navbar.tsx** — Responsive navigation header with active section tracking, theme toggle, and desktop/mobile actions.
- **src/components/HeroSection.tsx** — Executive banner with aligned status pills, core metrics, and contact action CTAs.
- **src/components/AiChatbotWidget.tsx** — Floating & header-triggered AI Assistant UI with Markdown formatting.
- **src/services/aiChatService.ts** — Client-side stream consumer + dynamic dossier fallback search engine.
- **src/data/resumeData.ts** — Typed master data module containing career history, metrics, and skills.
- **src/data/candidateDetailedDossier.ts** — Deep candidate background knowledge base supplying context to Gemini.

4. Technical Handover & Operation Commands

Command	Description / Action
`pnpm run dev`	Starts local Vite development server with HMR at http://localhost:5173/.
`pnpm run build`	Executes TypeScript type check (tsc -b) and builds production bundle in dist/.
`git push origin main`	Triggers automated Vercel CI/CD pipeline, building and deploying live to production.

5. Technical Recruiter & Engineering Manager Interview Q&A;

Below are exact technical questions engineering leaders and interviewers ask during executive screenings, accompanied by Kannan's articulated answers.

Q1: How did you achieve a 96% monorepo build acceleration at SuperOps?

Answer: I solo-led a 3-week migration from Webpack 5 to Rust-powered Rspack across 12 monorepo packages. By replacing JavaScript-based Webpack compilation with Rspack's parallelized C++/Rust architecture, cold-start build compilation dropped from 2 minutes (120s) down to 5 seconds (a 96% acceleration), with HMR updates under 50ms. This saved hundreds of engineering hours monthly.

Q2: How does the AI candidate assistant work technically under the hood?

Answer: The AI assistant uses HTTP Streamable Web Streams (fetch + ReadableStream) backed by Google Gemini 1.5 Flash. When a user asks a question, the client streams the query to a serverless proxy (api/chat.ts) which injects a detailed candidate dossier into the model's system prompt. Tokens are streamed back incrementally with TTFT under 80ms.

Q3: How do you handle API key security and LLM observability in production?

Answer: The Gemini API key is stored exclusively as a server-side environment variable (GEMINI_API_KEY) on Vercel and is never exposed to the client browser. For observability, I integrated Langfuse APM SDK on the serverless proxy, tracking generation latency, token consumption, and error states for every query.

Q4: What are Model Context Protocol (MCP) servers and write-scope safety guardrails?

Answer: MCP servers expose codebase AST symbols and component contracts directly to AI agents. To prevent autonomous agents from breaking production systems, I authored strict write-scope guardrails that allow agents to modify feature modules (src/features/**) while strictly locking down core schemas (src/core/**).

Q5: How does your AI test-authoring platform and self-healing suite function?

Answer: Using React and Node.js, I architected an automated end-to-end test platform where autonomous LLM agents generate Playwright test specs. If a UI release changes DOM selectors, the self-healing algorithm inspects DOM trees and updates selectors dynamically, maintaining a 232-spec regression suite in Jenkins CI.

Q6: What is your relocation readiness and notice period for Dubai, UAE?

Answer: I am ready to relocate immediately to Dubai, UAE upon completion of my 60-day notice period. I require employment visa sponsorship and plan to settle long-term in Dubai with my family.

Q7: What target roles are you evaluating in Dubai?

Answer: I am evaluating Senior Lead / Lead Software Engineer and Engineering Manager positions with product-led technology companies in Dubai, UAE.